Case Study

Analysing the Impact of Weather on Citibike Usage in NYC

Problem Context

New York City operates a bike-sharing service, Citibike. We're interested in understanding the impact of weather conditions on bike rental volume. For instance, how does a rainy or snowy day affect user behaviour compared to a clear, sunny day?

You are expected to explore the data, define the right questions, and guide the solution end-to-end — just like a consultant would. This is your opportunity to showcase not just your technical capabilities, but also your problem-solving mindset, communication skills, and business acumen.

This task is intentionally open-ended. You're encouraged to apply best practices, demonstrate initiative, and think critically about what matters.

The Task

You are required to:

- Design and implement an ELT pipeline using Snowflake and dbt Cloud.
- Ingest, transform, and model data from two distinct sources:
  - Citibike trip data
  - New York City weather data
- Deliver insights using Power BI, demonstrating any correlation between weather conditions and trip volume.

In addition, be prepared to present and walk us through your approach, including your assumptions, architectural decisions, and the rationale behind your insights — as you would in a client engagement.

Data Sources

- Citibike Trips:
  - Public portal: Citibike System Data
  - Direct S3 access via Snowflake external stage: s3://snowflake-workshop-lab/japan/citibike-trips/
- NYC Weather Data:
  - JSON format
  - Direct S3 access via Snowflake external stage: s3://snowflake-workshop-lab/weather-nyc

Tooling & Resources

You can download full trial versions of the software below:

- Snowflake: 30 day trial here
- dbt: here
- Power BI – here

You can also access training material in the links below:

- Snowflake: Snowflake Training Resources
- dbt: dbt Cloud Fundamentals
- Power BI: Power BI Learning Paths

What We're Looking For

This is not about perfection — we're more interested in your approach. We'll evaluate:

- **Architecture**: Is your ELT pipeline logical, modular, and scalable?
- **Data Modeling**: Can you translate raw data into structured, insightful business tables?
- **Data Integration**: How effectively do you merge and relate diverse datasets?
- **Assumptions & Communication**: Are your decisions well-reasoned and clearly explained?
- **Insight Delivery**: Is your Power BI dashboard clear, actionable, and story-driven?

Deliverables

There's no template for delivery, but your output might include:

- A working Snowflake environment with all tables/models
- A dbt Cloud project with documentation and tests
- A Power BI dashboard or story-driven report
- A showcase of your solution to our engineering team

Good Luck

We look forward to seeing your creativity and technical expertise in action. Good luck!